Linear Optimization (311.160)
Exercise sheet 2

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The course workflow says to solve the exercises on the sheet, but not the classroom task. I therefore leave classroom task 2.1 out of this PDF.

Exercise 2.1: standard form, feasibility and boundedness

I use the lecture's standard form

$$\max\ c^{\mathsf{T}}x \quad \text{subject to} \quad Ax = b, \quad x \geq 0.$$

For the original problem, introduce a slack variable for each inequality:

$$\begin{aligned} \max \quad & 2x_1 + x_2 \\ \text{subject to} \quad & 2x_1 - x_2 + s_1 = 12, \\ & x_1 + s_2 = 8, \\ & x_1, x_2, s_1, s_2 \geq 0. \end{aligned}$$

Thus, with $x = (x_1, x_2, s_1, s_2)^{\mathsf{T}}$,

$$c = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \qquad b = \begin{pmatrix} 12 \\ 8 \end{pmatrix}, \qquad A = \begin{pmatrix} 2 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

The point $(x_1, x_2, s_1, s_2) = (0, 0, 12, 8)$ is feasible. The LP is unbounded: for any $t \geq 0$, $(x_1, x_2) = (0, t)$ satisfies the original constraints and has objective value t. Hence the objective can be made arbitrarily large.

Exercise 2.2: reading the graph

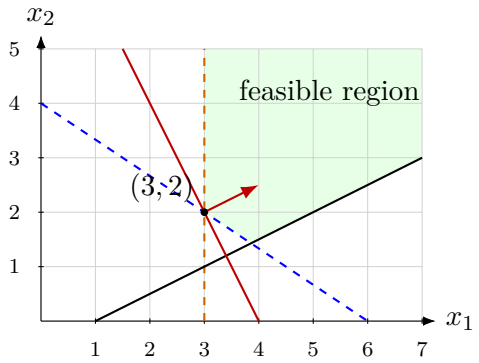
The shaded region lies to the right of $x_1 = 3$ and above both dashed/solid boundary lines. Reading their intercepts and slopes gives

$$2x_1 + 3x_2 \geq 12, \qquad x_1 - 2x_2 \leq 1, \qquad x_1 \geq 3.$$

The objective contours in the figure have slope -2 and are parallel to $2x_1 + x_2 = \text{constant}$. Their direction of increase points up and to the right, so the LP shown is

$$\begin{aligned} \max \quad & 2x_1 + x_2 \\ \text{subject to} \quad & 2x_1 + 3x_2 \geq 12, \\ & x_1 - 2x_2 \leq 1, \\ & x_1 \geq 3. \end{aligned}$$

A feasible ray is $(x_1, x_2) = (5 + t, 2 + t/2)$ for $t \geq 0$. It satisfies $x_1 - 2x_2 = 1$ and $2x_1 + 3x_2 = 16 + \frac{7}{2}t \geq 12$, while the objective is $12 + \frac{5}{2}t$. The objective therefore grows without bound.



Dashed blue: $2x_1 + 3x_2 = 12$; black: $x_1 - 2x_2 = 1$; orange: $x_1 = 3$; red: objective contour.

Exercise 2.3: graphical solution

The feasible region is in the first quadrant, below both resource constraints, and between $x_2 = 0$ and $x_2 = 4$. Its corner points are

$$(0, 0), \quad \left(\frac{16}{3}, 0\right), \quad (4, 4), \quad (0, 4).$$

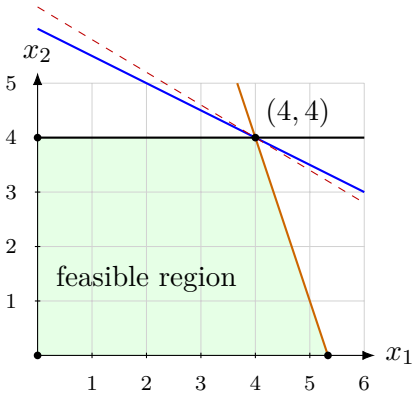
The two resource constraints meet at $(4, 4)$, which also lies on $x_2 = 4$. Evaluating the objective at the corners gives

(x_1, x_2)	$(0,0)$	$(16/3, 0)$	$(4,4)$	$(0,4)$
$3x_1 + 5x_2$	0	16	32	20

Thus $(x_1^*, x_2^*) = (4, 4)$ and the maximum value is 32. As a check, the constraints $3x_1 + x_2 \leq 16$ and $x_2 \leq 4$ imply

$$3x_1 + 5x_2 = (3x_1 + x_2) + 4x_2 \leq 16 + 16 = 32,$$

and $(4, 4)$ attains this bound.



Blue: $x_1 + 2x_2 = 12$; orange: $3x_1 + x_2 = 16$; black: $x_2 = 4$; red dashed: objective contour.